Confidence-gated correction of docking poses revisited: an independent reproduction shows that learned error reduction comes from ligand collapse

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Abstract

Learned post-docking correction can lower coordinate error, but an earlier study showed that it can also push acceptable poses out of tolerance and destroy chemical validity, and that a confidence gate only partly prevents this. We rebuilt that study end to end from public sources: 16,965 PDBbind/LP-PDBBind/MISATO complexes in a leakage-controlled manifest, the 428-complex PoseBusters set as an external cohort, Vina docking, the original model and gate, and a full PoseBusters audit. The central claims reproduce, more strongly: ungated correction reduced pooled squared error by 18.16% (internal) and 11.26% (external), yet lowered 2 Å success by -16.4 and -19.2 percentage points and left only 0.60% and 0.28% of poses chemically valid. New analyses explain why. Corrected ligands contract to a median 0.58 of their docked radius of gyration, and 96% of each correction is internal deformation; restricting the correction to a rigid motion preserves validity but removes all error reduction. Pooled error falls only because a few badly docked complexes dominate it; for the median complex it rises. The gate admitted almost nothing, failed its own correlation check on all three alternative splits, and the reported uncertainty shrinkage under distribution shift did not reproduce. 17.8% of external complexes are near-identical in sequence to training proteins. Per-atom regression under a squared-error objective learns to collapse ligands; pose refinement should be evaluated per complex and constrained to chemically valid motions.

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Keywords: Molecular docking, pose refinement, reproducibility, uncertainty estimation, confidence gating, chemical validity, PoseBusters, dataset shift

1. Introduction

Molecular docking predicts a bound conformation of a ligand in a protein binding site and is used to prioritise compounds and to form binding-mode hypotheses (Trott and Olson, 2010; Eberhardt et al., 2021; Pagadala et al., 2017). Learned post-docking correction is an attractive complement: a model that sees the docked ligand and its pocket can predict a residual displacement towards the experimentally observed pose. The risk is asymmetric. A correction can rescue a poor pose, but the same operation can move an acceptable pose out of tolerance, and it can do so while producing a molecule whose bond lengths, angles or contacts are physically implausible—a failure that root-mean-square deviation (RMSD) cannot see (Buttenschoen et al., 2024).

Mwamughunda et al. (2026) studied this risk with a confidence-gated correction model: a ligand–pocket cross-attention network with a coordinate-correction head and a heteroscedastic log-variance head, a confidence gate calibrated on validation data only, and an evaluation on three axes (continuous error, RMSD-threshold transitions, and a PoseBusters chemical-validity audit). Their central findings were that ungated correction reduces mean squared error while lowering or leaving unchanged RMSD-threshold success, that it collapses chemical validity, and that gating recovers validity only partially, less so on an externally shifted cohort.

These conclusions matter beyond one model, because they concern how learned refinement should be evaluated. They also rest on one dataset construction, one split, and a gate whose threshold is an absolute uncertainty value. This paper therefore asks two questions. **(Q1) Reproducibility:** do the original findings hold when the full pipeline is rebuilt independently from public sources, using the same datasets and protocol? **(Q2) Gaps:** can the failure modes the original study diagnosed—validity loss, miscalibration under shift, and single-split evidence—be measured more precisely or mitigated with label-free changes to the decision rule?

Our contributions are:

1. An independent, end-to-end reconstruction of the study from public data: the PDBbind v2020 P-L, LP-PDBBind and MISATO identifier

intersection, an external PoseBusters cohort, AutoDock Vina docking, four candidate pocket encoders, the three-stage gate, and a full-cohort PoseBusters audit, with every deviation reported (Section 3.8).

2. A side-by-side comparison of every headline number of the original study with the reproduction (Table 1).
3. Seven new analyses addressing gaps identified in the original work (G1–G7): a quantile-matched gate that adapts to uncertainty shift without labels, a validity-veto gate, rigid-body projection of the correction, deep-ensemble uncertainty, sequence-level leakage between the training and external cohorts, a diagnosis of the uncertainty shift, and split-to-split variance over three alternative splits.
4. A public code release that regenerates every number and figure in this paper.

2. Related work

Classical docking balances conformational search and scoring (Meng et al., 1992; Morris et al., 2009; Trott and Olson, 2010); AutoDock Vina 1.2 added a Python interface used here (Eberhardt et al., 2021). PDBbind is the standard source of characterised complexes (Wang et al., 2004, 2005); LP-PDBBind reorganises it to reduce train–test leakage (Li et al., 2023), and MISATO adds molecular-dynamics trajectories for most PDBbind complexes (Siebenmorgen et al., 2024). Learned docking methods predict poses directly or generatively (Stärk et al., 2022; Corso et al., 2023; Pei et al., 2023), and general structure predictors now model complexes (Jumper et al., 2021; Abramson et al., 2024). PoseBusters showed that many learned methods produce poses that pass RMSD criteria but fail basic physical checks, and that performance drops on proteins dissimilar to the training data (Buttenschoen et al., 2024). Later benchmarks report similar gaps between geometric accuracy and physical or interaction validity (Morehead et al., 2026; Errington et al., 2025; Li et al., 2025; Gu et al., 2025).

Heteroscedastic regression models input-dependent noise through a learned variance (Kendall and Gal, 2017); deep ensembles estimate uncertainty from model disagreement (Lakshminarayanan et al., 2017); calibration methods map scores to reliable probabilities (Guo et al., 2017; Kuleshov et al., 2018). Uncertainty estimates degrade under dataset shift (Ovadia et al., 2019; Quiñonero-Candela et al., 2009), including for protein–ligand affinity (Rayka and Naghavi, 2025). Our quantile-matched gate borrows the rank-based view of conformal

prediction (Vovk et al., 2005): it fixes the fraction of cases admitted rather than an absolute score.

3. Materials and methods

The pipeline (Fig. 1) follows the original protocol stage by stage. All stages are deterministic given the released manifest and fixed seeds.

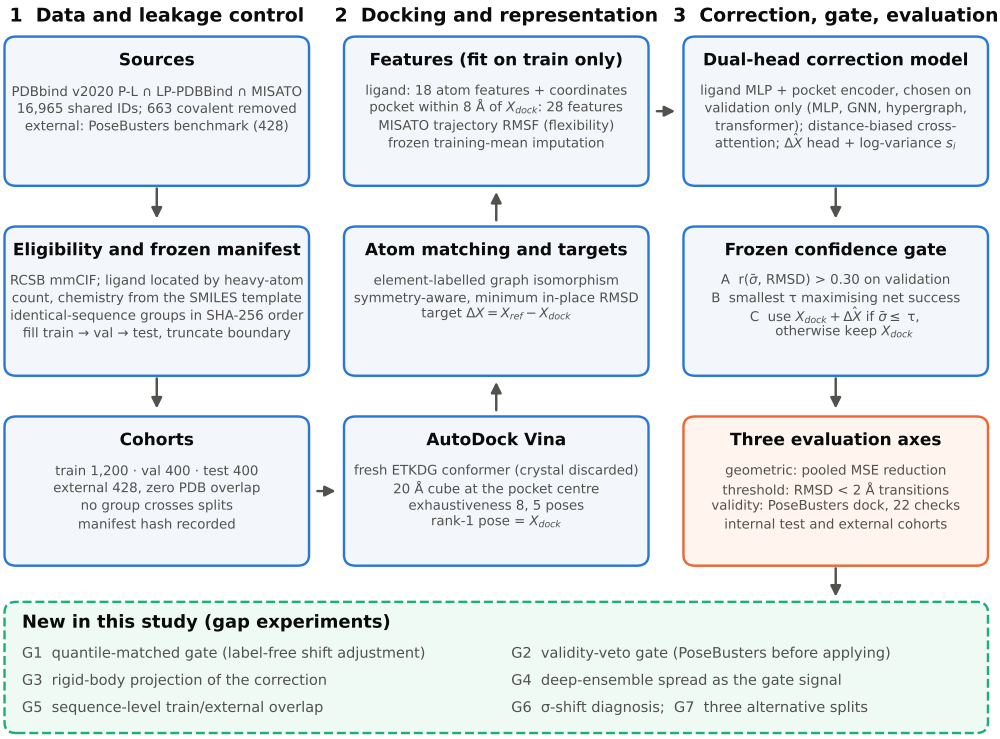


Figure 1: Methodology. Column 1 builds the leakage-controlled cohorts, column 2 docks each complex and builds its representation, column 3 trains the dual-head correction model, applies the frozen confidence gate and evaluates every pose on three axes. The band lists the new analyses G1–G7 (G8, the deformation analysis, is in Section 4).

3.1. Datasets and eligibility

Internal pool. The PDBbind v2020 P-L set (19,443 complexes, as distributed in LP-PDBBind) was intersected with the MISATO MD identifier lists (train,

validation and test files; 16,972 identifiers), giving 16,965 shared complexes (original study: 16,627). 663 covalent complexes were excluded using the LP-PDBBind flag. Structures were downloaded from the RCSB PDB (Berman et al., 2000) as mmCIF. The crystal ligand was located as the single non-polymer residue whose heavy-atom count equals that of the LP-PDBBind SMILES and whose coordinates admit the SMILES bond orders (RDKit `AssignBondOrdersFromTemplate`); the ligand was then sanitised with RDKit. Ligands spanning several residues (mostly peptides) cannot be located this way and were excluded (357 complexes), as were 28 entries whose structures failed to parse. The receptor retains polymer residues, metal ions and organic cofactors, and drops water, common crystallisation additives, hydrogens, the ligand, and any hetero group within 1 Å of a ligand atom (alternate ligand copies).

External cohort. The 428-complex PoseBusters benchmark set (Buttenschoen et al., 2024), with its supplied receptor and ligand files. Internal–external PDB-identifier overlap is zero.

Leakage-controlled manifest. Complexes were grouped by identical protein sequence (LP-PDBBind chain sequence). Groups were ordered by the SHA-256 digest of the group identifier and used to fill the training (1,200), validation (400) and internal-test (400) splits in that order; only the boundary group is truncated and its unselected members retired. Eligibility was checked before a complex took a slot. The manifest (SHA-256 recorded) was frozen before any modelling. No group crosses splits.

3.2. Docking and atom matching

For each complex a fresh conformer was generated from the ligand graph with ETKDG and MMFF94 optimisation (Riniker and Landrum, 2015; Halgren, 1996), discarding the crystal coordinates, and prepared with Meeko. Receptors were converted to PDBQT with Open Babel (O'Boyle et al., 2011); receptor atoms of types Vina cannot score were removed. AutoDock Vina was run with a 20 Å cubic box at the crystal-ligand centroid (pocket-informed), exhaustiveness 8, five poses and a fixed seed; the rank-1 pose is the baseline X_{dock}. Docked and reference heavy atoms were matched by element-labelled graph isomorphism, choosing among graph automorphisms the correspondence with the smallest in-place RMSD (symmetry-corrected RMSD, no superposition). Complexes whose graphs did not match were excluded.

3.3. Pocket representation and MISATO flexibility

The pocket is the set of residues with any heavy atom within 8 Å of a docked ligand heavy atom, so it is defined from the docked pose alone. Each residue node has 28 features: a 20-way amino-acid one-hot encoding, the Cα position and side-chain-centroid offset relative to the docked-ligand centroid, a local degree (Cα neighbours within 10 Å), and a flexibility value. Ligand atoms carry 18 chemistry features (element, degree, aromaticity, ring membership, formal charge, hydrogen count, hybridisation) and their coordinates. Edges link Cα pairs within 10 Å; hyperedges are chain segments of consecutive residues.

Flexibility is the per-residue root-mean-square fluctuation (RMSF) of the MISATO trajectory. Because the 133 GB MISATO file could not be stored, each complex's trajectory (100 frames) was read remotely by HTTP range requests. MISATO coordinates are expressed in a different frame and residue numbering, so frame 0 was registered onto the crystal by matching the ligand's heavy-atom graph (element-labelled isomorphism) and a Kabsch fit (Kabsch, 1976); frames were aligned on protein heavy atoms before computing per-atom RMSF, and each pocket residue receives the mean RMSF of MISATO atoms within 1.5 Å of its heavy atoms. Registrations with ligand RMSD above 2 Å or median protein nearest-neighbour distance above 2.5 Å were rejected. Residues without a mapped atom, complexes with rejected registrations, and all external complexes receive the frozen training-set mean, the original study's prespecified imputation rule. Continuous features and the coordinate scale were fitted on training complexes only.

3.4. Correction model and uncertainty

For N matched atoms the target is $\Delta X = X_{ref} - X_{dock}$, the model predicts $\hat{\Delta X} = f_\theta(X_{dock}, G_{pocket})$, and the corrected pose is $X_{corr} = X_{dock} + \hat{\Delta X}$. The common architecture has a ligand MLP encoder, a pocket encoder, distance-biased cross-attention from ligand atoms to pocket residues, a three-coordinate correction head and a per-atom log-variance head s_i (clamped to $[-12, 12]$), with $\sigma_i = \exp(s_i/2)$. The training objective is the heteroscedastic negative log-likelihood

$$\mathcal{L}_{NLL} = \tfrac{1}{2}\,\mathrm{mean}_i\Big[3s_i + \|\hat{\Delta X}_i - \Delta X_i\|_2^2 / e^{s_i}\Big]. \tag{1}$$

The four candidate pocket encoders are an MLP over residue features, a distance-weighted message-passing GNN (Gilmer et al., 2017), a residue-incidence hypergraph encoder with flexibility-weighted segment attention,

and a transformer with a learned distance bias. All use hidden size 64 and two encoder layers, and are trained with Adam (learning rate 10^{-3}), at most 15 epochs and patience 5 on validation NLL, with random proper rotations of all coordinates and targets during training only. Five seeds (0, 42, 123, 999, 2026) were trained per architecture; the architecture with the highest mean validation pooled MSE reduction was selected. Held-out data were not used for selection.

3.5. Confidence gate

Let $\bar{\sigma}_j$ be the mean predicted σ of seed–complex row j. Stage A requires the pooled validation Pearson correlation between $\bar{\sigma}$ and baseline RMSD to exceed 0.30. Stage B sweeps every distinct validation $\bar{\sigma}$ and selects the smallest threshold maximising pooled validation net success,

$$\tau = \min\{t : \mathrm{NetSuccess}(t) = \max_{t'} \mathrm{NetSuccess}(t')\}, \tag{2}$$

where NetSuccess is the RMSD < 2 Å success rate when the correction is applied to rows with $\bar{\sigma}_j \leq t$ minus the baseline success rate. Stage C applies the frozen τ to held-out data without adjustment.

3.6. Evaluation

Geometric: pooled MSE reduction $100\,(\mathrm{MSD}_{base} - \mathrm{MSD}_{corr})/\mathrm{MSD}_{base}$ over all atoms. *Threshold:* success is RMSD < 2 Å; transitions are preservation (success→success), harm (success→failure), rescue (failure→success) and persistent failure. *Validity:* the full PoseBusters `dock` configuration (22 binary checks: loading, sanitisation, InChI, connectivity, radicals, bond lengths and angles, internal clash, ring and double-bond flatness, internal energy, and protein-ligand distance and volume-overlap checks), applied to heavy-atom poses against the receptor; a pose is valid only if every check passes, and a check not computed after an upstream failure counts as failed. Gated poses are the corrected pose when the gate fires, otherwise the baseline. Uncertainty intervals are reported in two ways: a Student-t interval over the five seed-level rates, and a 10,000-resample paired complex-level percentile bootstrap.

Statistical validation. A training-only permutation test relates each atom's distance to the nearest pocket Cα to its target displacement magnitude, shuffling magnitudes within complexes (10,000 permutations). A rotational

audit compares predictions before and after a random rotation (rotated back) for ten validation complexes and five seeds. A synthetic identity check verifies $\mathrm{RMSE}_{component} = \mathrm{RMSD}/\sqrt{3}$.

3.7. New analyses (G1–G7)

G1 Quantile-matched gate. The validation admit fraction q is frozen; each held-out cohort is gated at the q-quantile of its own $\bar{\sigma}$ distribution. No held-out label is used. This tests whether matching the admit rate, rather than the absolute threshold, removes the effect of uncertainty-scale shift.

G2 Validity veto. A correction is applied only if the corrected pose passes all PoseBusters checks (alone, and combined with the σ gate). PoseBusters needs only the pose and receptor, so the rule is deployable.

G3 Rigid-body projection. The free per-atom correction is replaced by the rigid motion of the docked pose that best fits it (Kabsch). Internal geometry is preserved by construction; the gate is recalibrated on validation with the same rule.

G4 Deep-ensemble uncertainty. The five seeds' mean correction is gated by their spread (mean per-atom standard deviation), calibrated exactly as in Stage B; a rigid-projected variant is also evaluated.

G5 Sequence-level leakage. Exact and near-identical (3-mer Jaccard $\geq$ 0.9) chain-sequence overlap between the training set and the external cohort, and outcomes stratified by overlap.

G6 Uncertainty-shift diagnosis. Per-cohort $\bar{\sigma}$ distributions (two-sample Kolmogorov–Smirnov tests), σ–RMSD correlation per cohort, and cohort descriptors (ligand size, pocket size).

G7 Split variance. The docked internal pool is re-partitioned three times with the same group-aware rule and different hash salts; the selected architecture is retrained (five seeds each) and re-evaluated.

3.8. Deviations from the original protocol

- *Structures:* PDBbind's curated files require registration, so structures came from the RCSB PDB and ligand chemistry from LP-PDBBind SMILES. Multi-residue ligands were excluded (the original study excluded none for this reason).

- *Grouping:* identical protein sequence was used as the leakage group; the original used LP-PDBBind grouping (1,792 candidate-touching groups versus 1,581 here).
- *Software:* AutoDock Vina 1.2.7 (original: 1.2.0), RDKit 2026.09, PoseBusters 0.6.5.
- *Docked-pose export:* Meeko's RDKit export was used instead of an Open Babel PDBQT→SDF conversion; the Open Babel conversion of the same pose was retained to test the conversion artifact reported in the original study.
- *Flexibility:* MISATO RMSF obtained through the registration procedure above; the original's exact mapping was not specified.
- *Validity audit:* poses were audited as heavy-atom molecules for all variants, which avoids hydrogens being left behind by a heavy-atom correction.

4. Results

4.1. Cohorts and docking baseline

All 400 internal-test and 428 external complexes docked and matched (no atom-matching exclusions), giving 2,000 and 2,140 seed–complex rows. Vina rank-1 success (RMSD < 2 Å) was 41.5% internally and 54.4% externally, against 55.6% and 60.7% in the original study (Table 1). The internal gap is the largest difference in inputs between the two studies and is consistent with the use of RCSB rather than PDBbind-prepared structures; mean baseline RMSD was 3.78, 3.78 and 3.15 Å for validation, internal test and external.

4.2. Architecture selection

On validation only, the four encoders reached almost identical pooled MSE reductions (Fig. 2): MLP 16.25%, GNN 16.22%, hypergraph 16.17% and transformer 16.64%. The transformer had the highest mean and was selected; paired seed-level t-tests favoured it over each alternative (MLP $p = 0.001$, GNN $p = 0.012$, hypergraph $p = 0.007$), but the one-way ANOVA across all four was not significant ($F = 2.90$, $p = 0.067$). The original study selected the MLP. In both studies the spread between encoders is under one

percentage point, so the choice of encoder is not what drives the outcomes below.

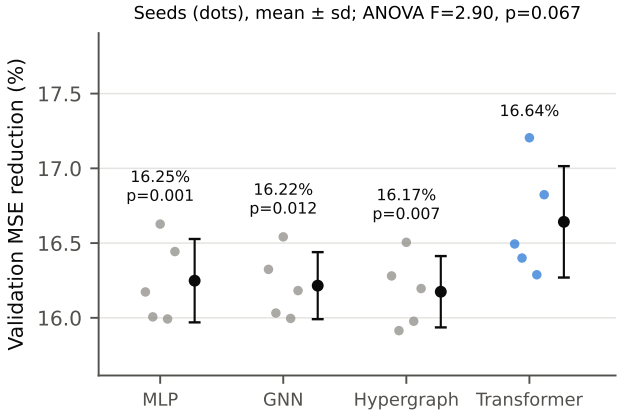


Figure 2: Validation-only architecture selection: mean pooled MSE reduction over five seeds (dots: seeds).

4.3. Statistical validation

The training-only permutation test (1,200 complexes, 31,706 heavy atoms, 10,000 permutations) depends on how "pocket distance" is defined, which the original study did not specify. Using the distance from each atom to the pocket centroid, the correlation with target displacement was $r = 0.267$ against a within-complex null mean of 0.089 ($p = 1.0e{-}04$, two-sided around the null mean). Using the distance to the nearest pocket Cα, it was $r = 0.063$, significantly *below* its null mean of 0.095 ($p = 1.0e - 04$). Because within-complex shuffling preserves the between-complex association, the null is not centred at zero; a test that compares $|r|$ against $|r_{null}|$ would report $p = 1$ for the second definition. The rotational audit (50 observations) gave a mean prediction discrepancy of 0.68 Å (median 0.67, 95th percentile 1.15, maximum 1.34 Å), confirming that the cross-attention model is not rotation equivariant (original: 0.53 Å mean). The metric identity held to 2.2e-16.

4.4. RMSD-threshold outcomes

Ungated correction reduced pooled MSE by 18.16% internally (seed range 17.85–18.87%) and 11.26% externally (8.02–13.10%), but it lowered success from 41.5% to 25.1% internally (-16.4 pp; bootstrap 95% CI [-19.5, -13.2]) and from 54.4% to 35.3% externally (-19.2 pp; CI [-22.4, -16.0]; Fig. 3). Both

decreases are significant and much larger than the original internal decrease (−3.6 pp), and the original external change was null. Harm dominated rescue among the transitions (Fig. 3): internally 329 harmed versus 2 rescued rows, externally 415 versus 5.

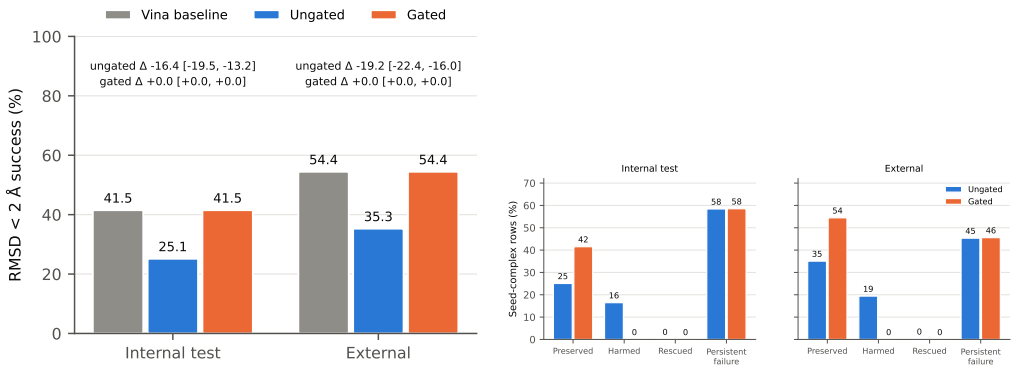


Figure 3: Left: RMSD < 2 Å success of the Vina baseline, ungated and gated correction, with net change and bootstrap 95% CI. Right: transition classes (share of seed–complex rows).

4.5. Confidence gate

Stage A passed narrowly: the validation correlation between mean σ and baseline RMSD was 0.376 (threshold 0.30). Stage B selected $\tau = 1.573$, the smallest threshold reaching the maximum validation net success of 0.10 pp—the same value the original study reported—but this threshold admits only 2.9% of validation rows, 2.7% of internal-test rows and 1.3% of external rows (original: 24.15%, 23.4% and 55.5%). The gated system is therefore almost identical to the baseline: gated success equalled baseline success in both cohorts (net +0.0 and +0.0 pp). The baseline-RMSD control gate reproduced the original observation that no "correct only good poses" threshold helps (best net 0.00 pp), but the opposite rule—correct only poses with baseline RMSD ≥ 2.06 Å— reached 0.15 pp on validation, so on this split a baseline-quality signal (if it were available at inference) would outperform the learned uncertainty.

4.6. Chemical validity

Baseline poses passed all 22 PoseBusters checks in 98.50% (internal) and 98.13% (external) of rows. After ungated correction only 0.60% and 0.28%

passed (Fig. 4); among rows with a valid baseline, 99.39% and 99.71% became invalid. The gate kept 96.30% and 97.10% valid (newly invalid: 2.23% and 1.05%; relative reduction 97.8% and 98.9%), but only because it rarely fired. Validity loss was independent of the RMSD transition class: even among *preserved* rows, 98.0% (internal) and 99.3% (external) of corrected poses were invalid (Fig. 5). The failing checks were intramolecular (Fig. 5): bond lengths (99.2% of corrected internal poses), bond angles (98.5%), internal steric clash (94.9%) and internal energy (89.4%); protein-distance failures were far less common (22.9%). Among 96 preserved-but-invalid internal poses (seed 0) the shortest bond shrank to a median 49% of its docked length (minimum 29%), and bond angles changed by up to 46°.

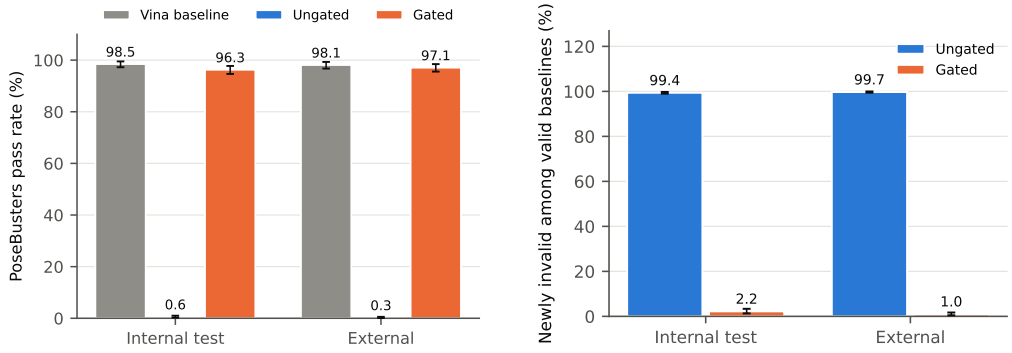


Figure 4: Left: PoseBusters pass rate (bootstrap 95% CI). Right: rate at which correction invalidates a valid baseline.

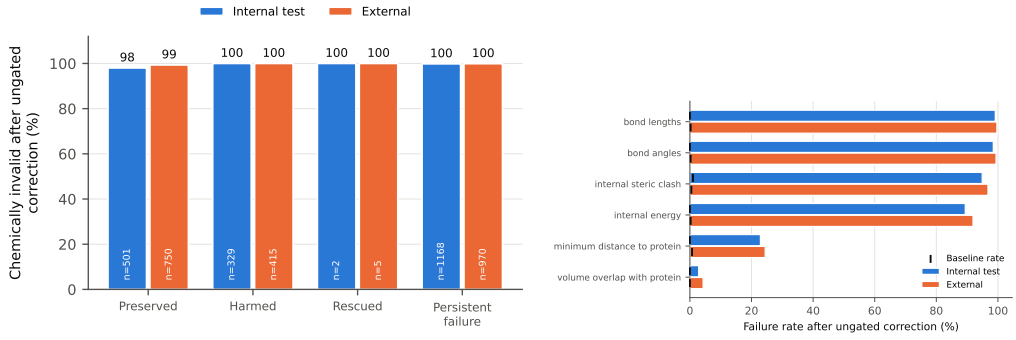


Figure 5: Left: invalidity of corrected poses within each RMSD transition class. Right: failure rate of each PoseBusters check after ungated correction (bars) against the baseline (ticks); checks below 1% are omitted.

The conversion artifact. The original study traced many baseline sanitisation failures to nitrogen atoms with valence four and no formal charge introduced by PDBQT→SDF conversion. Converting the same rank-1 poses with Open Babel reproduced it: 111 internal-test complexes (128 such atoms) and 5 external complexes failed sanitisation, close to the original 150 and 4. Meeko's export, used for all analyses here, does not produce the artifact, so no repair was needed.

4.7. Why correction destroys chemistry (G3, G8)

Every predicted correction was decomposed into the rigid motion that best maps the docked pose onto the corrected pose and a residual internal deformation. The internal part made up a median 96% (internal) and 96% (external) of the correction's squared magnitude. Corrected ligands contracted to a median 0.58 (interquartile range 0.53–0.65) and 0.56 of their docked radius of gyration, and 97.3% of internal rows shrank below 0.8, whereas crystal poses keep the docked size (median ratio 1.00; Fig. 6a). When the correction was replaced by its rigid part (G3), validity was largely preserved (89.5% and 84.0% passed), but the error reduction vanished (-0.09% and -1.13%), and the recalibrated gate admitted nothing. The pooled error reduction is therefore produced by contracting the ligand, not by moving it towards the crystal pose.

Pooled MSE also hides who benefits. Contraction reduces the error of badly docked poses, whose atoms are far from the reference in all directions, but increases it for good poses (Fig. 6b). The median per-complex change in squared error was -8.3% internally and -74.7% externally—an *increase*—while the pooled reduction was positive because a minority of high-error complexes dominates the pooled sum.

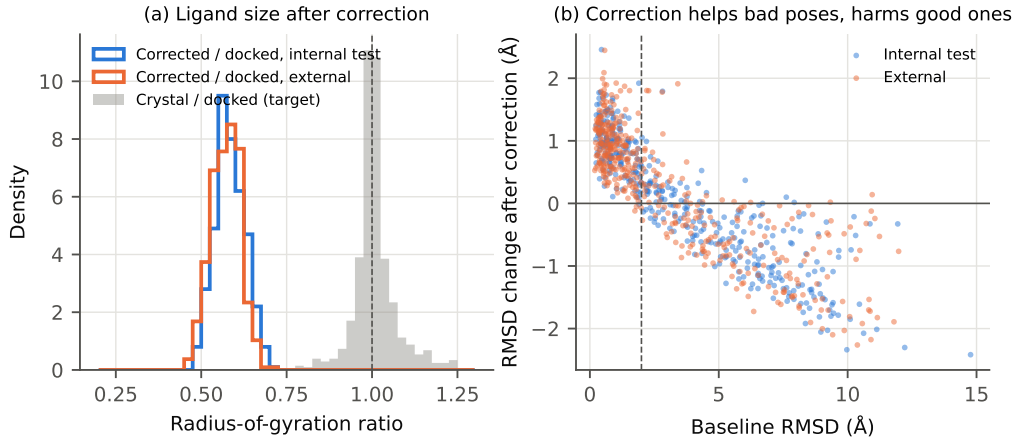


Figure 6: (a) Radius-of-gyration ratio of corrected to docked ligands (lines) and of crystal to docked ligands (shaded). (b) RMSD change after correction against baseline RMSD (seed 0); dashed line: 2 Å.

4.8. Alternative decision rules (G1, G2, G4)

None of the label-free decision rules produced a positive held-out success change (Fig. 7). The quantile-matched gate (G1) fixed the admit fraction at the validation value (2.9%) in each cohort, removing the absolute-scale dependence, but success did not change (net +0.00 and +0.00 pp) and external validity fell slightly (95.4% against 97.1% for the frozen gate). The validity veto (G2) is deployable—PoseBusters needs no ground truth—but almost no corrected pose passed it (admit 0.6% and 0.3%), so it reduces to "never correct". The five-seed ensemble (G4) reduced pooled MSE slightly more than single models (19.06% and 12.59%) and its spread correlated with baseline RMSD ($r = 0.260$, below the 0.30 Stage A threshold), but averaging five contracting predictions produced invalid poses throughout (0.0% valid). A rigid ensemble kept validity (98.5%) with negligible geometric effect.

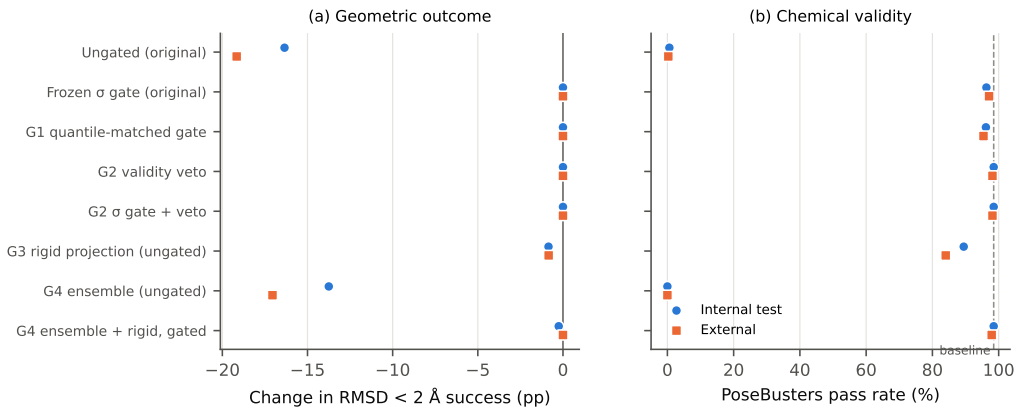


Figure 7: Held-out success change against PoseBusters pass rate for the original gate and the new decision rules.

4.9. Sequence-level leakage (G5)

Although no PDB identifier is shared, 8 external complexes contain a protein chain identical to a training chain and 76 (17.8%) are near-identical (3-mer Jaccard ≥ 0.9). On these 76 complexes ungated correction reduced pooled MSE by 18.97%, against 8.92% on the 352 novel ones; RMSD-threshold harm was large in both groups (-23.9 and -18.1 pp). PDB-identifier disjointness therefore overstates how "external" the PoseBusters cohort is for a PDBbind-trained model, and the external error reduction is concentrated on familiar proteins.

4.10. Uncertainty shift (G6)

The original study attributed weaker external gating to predicted σ being about 32% lower externally. Here the mean external σ differed from validation by only +1.1% (Kolmogorov–Smirnov $D = 0.043$, $p = 0.041$), while the internal test differed by -4.0% ($D = 0.113$). The σ–RMSD correlation fell from 0.376 (validation) to 0.309 (internal) and 0.273 (external), and σ was essentially unrelated to the per-complex benefit of correcting (Spearman 0.090, 0.068, -0.040). The gate fails here not because σ shifts but because it ranks poses by how wrong they are, not by whether correcting them helps (Fig. 8).

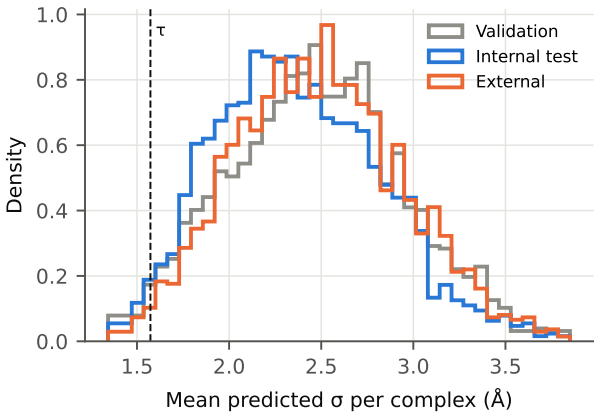


Figure 8: Distribution of mean predicted σ per complex by cohort; dashed line: frozen threshold τ.

4.11. Split-to-split variance (G7)

Re-partitioning the internal pool three times with the same group-aware rule (Fig. 9) left the main pattern intact: internal MSE reductions of 15.23, 14.30 and 14.34%, success changes of -17.3, -15.3 and -21.1 pp, and ungated validity below 1%. The gate, however, depends on the split: Stage A fails ($r = 0.262$), fails ($r = 0.286$) and fails ($r = 0.295$) on the three alternatives, so on every alternative split no gate would have been deployed at all.

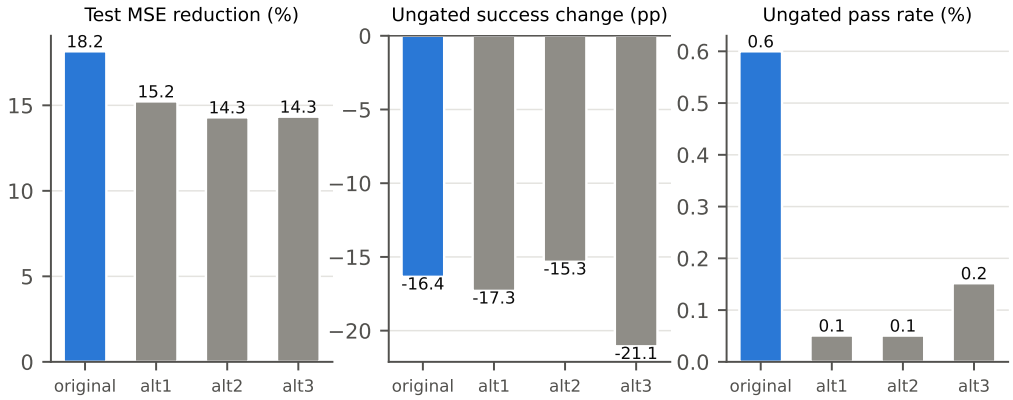


Figure 9: Original split and three alternative group-aware splits.

4.12. Comparison with the original study

Table 1: Headline results of the original study and of this reproduction (internal / external where two values are given).

Quantity	Original	Reproduction	Verdict
ID intersection	16,627	16,965	reproduced
Covalent exclusions	651	663	reproduced
Vina success (%)	55.6 / 60.7	41.5 / 54.4	partly
Selected encoder	MLP ($p_{\mathrm{ANOVA}} = 0.010$)	Transformer ($p_{\mathrm{ANOVA}} = 0.067$)	not reproduced
Pooled MSE reduction (%)	9.75 / 11.35	18.16 / 11.26	reproduced
Ungated success change (pp)	−3.6 / n.s.	-16.4 / -19.2	partly
Gate val. net success (pp)	+0.10	+0.10	reproduced
Gate admit rate val / int / ext (%)	24.2 / 23.4 / 55.5	2.9 / 2.7 / 1.3	not reproduced
External vs. validation σ	−32%	+1.1%	not reproduced
PoseBusters pass, baseline (%)	96.45 / 97.66	98.50 / 98.13	reproduced
PoseBusters pass, ungated (%)	31.22 / 59.02	0.60 / 0.28	partly
PoseBusters pass, gated (%)	89.04 / 83.64	96.30 / 97.10	partly
Newly invalid, ungated (%)	67.6 / 39.9	99.39 / 99.71	partly
Invalid within preserved class (%)	70.1 / 34.8	98.0 / 99.3	partly
Permutation r	0.51	0.267 (centroid)	partly
Rotation discrepancy, mean (Å)	0.53	0.68	reproduced
Open Babel artifact (complexes)	150 / 4	111 / 5	reproduced

"Partly" marks results with the same direction and conclusion but a clearly different magnitude.

5. Discussion

Interpretation of findings. The reproduction confirms the qualitative claims of the original study—continuous error falls while RMSD-threshold success and chemical validity fall with it—and shows that the effect is stronger and has a simple cause. The learned correction mostly contracts the ligand. Contraction lowers squared error for badly docked poses and for the pooled metric, but it breaks bonds and angles in nearly every pose and moves good poses out of tolerance.

Relation to the research questions. For Q1, the data pipeline, docking baseline, metric behaviour, rotation audit and conversion artifact reproduced closely; the selected encoder, the size of the RMSD harm, the gate's operating point and the uncertainty-shift explanation did not (Table 1). For Q2, measuring the gaps was more productive than closing them: none of the label-free decision rules made correction beneficial, but the analyses located the failure in the correction itself rather than in the gate.

Relation to prior work. PoseBusters showed that learned docking can satisfy RMSD criteria with invalid chemistry (Buttenschoen et al., 2024); the original study extended this to post-hoc correction (Mwamughunda et al., 2026). Our results add the mechanism: the conditional mean of atom positions under pose uncertainty lies inside the molecule, so a model trained with squared error (or a Gaussian likelihood, which shares its mean) learns to predict a contracted pose. This mirrors the regression-to-the-mean blurring familiar from deterministic structure predictors and is one reason generative docking methods sample poses instead of regressing them (Corso et al., 2023). The external sequence overlap agrees with PoseBusters' observation that performance depends on similarity to the training proteins.

Explanation of unexpected results. Three differences from the original need explanation. First, the harm is larger here, plausibly because our docking baseline is worse (larger targets, more contraction) and because our encoder choice was not the original MLP; the effect persisted across all four encoders and three alternative splits, so it is not an artefact of one model. Second, the gate admitted almost nothing: with stronger harm, the validation net-success curve peaks at a very low admit fraction. Third, the uncertainty did not shrink externally, so the original explanation of weaker external gating does not transfer; the deeper problem is that σ tracks how wrong a pose is, not whether correcting it helps.

Theoretical implications. Evaluating refinement by pooled MSE rewards contraction. A refinement objective should be defined over chemically valid poses—rigid-body motions, torsions, or a learned distribution over conformers—so that the optimum cannot leave the space of molecules. The decomposition into rigid and internal parts is a simple diagnostic that any coordinate-regression model can report.

Practical implications. For practitioners, a learned per-atom correction of this kind should not be applied to docking output without a validity check, and a validity veto is cheap and deployable. Gating on predicted uncertainty is not enough: on every alternative split the gate failed its own correlation check, and on the original split it chose to correct almost nothing. Reports of external performance should include sequence-level similarity to the training data, not only identifier overlap.

Limitations. Structures came from the RCSB PDB because PDBbind's curated files require registration; multi-residue ligands were therefore excluded and the internal docking baseline is weaker than in the original study. MISATO flexibility was mapped through our own registration because the original mapping was not specified. We ran one correction family (the original architecture and objective); other objectives may behave differently. Audits were performed on heavy-atom poses. Computing resources were a 4-core CPU without GPU, so seeds and splits were limited to those reported.

Future research. Natural next steps are correction models constrained to rigid and torsional motions, objectives that penalise bond-length and angle deviations, generative refinement that samples rather than averages, and gates trained directly on the per-complex benefit of correcting rather than on predicted error.

6. Conclusion

Learned post-docking correction promises to fix docking errors, but an earlier study warned that it can also break poses and chemistry, and that a confidence gate only partly helps. We rebuilt that study independently from public data to test its claims and the gaps it left open.

The main warnings hold, and more strongly. Applying the correction to every pose lowered pooled coordinate error but reduced docking success by about 16–19 percentage points and left fewer than 1% of poses chemically valid.

Our new analyses show why. The model learns to shrink each ligand towards its centre: this lowers the average error, because a few badly docked poses dominate it, but distorts bonds and harms good poses. Moving the ligand only as a rigid body keeps the chemistry valid but removes the benefit.

The confidence gate did not solve this. It rarely corrected anything, failed its own check on all three alternative splits, and the reported uncertainty shift under distribution change did not reproduce.

The takeaway is that pose refinement should be judged per complex, on validity as well as geometry, and built so that it can only produce real molecules.

Data and code availability

All code, the frozen manifest and its hash, every result file and the scripts that regenerate each number (`numbers.tex`) and figure in this paper are in the `DockingPoseCorrection` folder of the accompanying repository. Source data are public: LP-PDBBind, the MISATO identifier lists and trajectories (Zenodo record 7711953), the RCSB PDB, and the PoseBusters benchmark (Zenodo record 8278563).

Declaration of generative AI use

The authors used an AI coding assistant (Claude, Anthropic) to implement the reproduction pipeline, run the experiments and draft the manuscript. The authors reviewed and are responsible for all content.

Declaration of competing interest

The authors declare no competing interests.

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